

Impact of Health Indicators on Economic Growth: A Panel Data Analysis of South Asian and South East Asian Economies from the era of high economic growth

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1. Introduction

Health and economic growth are closely interconnected. Improvements in health outcomes enhance labour productivity, increase human capital formation, and contribute to long-term economic development. Healthy populations are more capable of participating effectively in economic activities, while higher income levels enable greater investment in healthcare services and public welfare. Consequently, the relationship between health and economic growth has attracted significant attention in development economics.

South Asia and Southeast Asia together account for a substantial share of the world's population and include economies at different stages of development. The regions comprise both developing and emerging economies that have experienced rapid economic transformation during the twenty-first century. Countries such as India, Bangladesh, Indonesia, Vietnam, and Cambodia have recorded significant improvements in income levels and social indicators, while advanced economies such as Singapore and Brunei have achieved relatively high standards of living and healthcare outcomes. Despite this progress, considerable disparities continue to exist across countries in terms of healthcare expenditure, life expectancy, child mortality, and economic performance.

During the period 2000–2020, both regions witnessed notable improvements in health indicators. Life expectancy increased steadily, under-five mortality rates declined, and healthcare expenditure expanded in many countries. At the same time, economic growth remained an important policy objective as governments sought to improve living standards and accelerate development. Given the complementary relationship between health and economic development, it is important to examine whether improvements in health outcomes and healthcare investment contribute significantly to economic growth.

2. Hypothesis

This study examines the relationship between health indicators and economic growth in selected South Asian and Southeast Asian countries using panel data analysis.

The study uses Gross Domestic Product per capita as the dependent variable, while life expectancy, current health expenditure per capita, domestic private health expenditure per capita, government expenditure, and under-5 mortality rate are used as explanatory variables.

The hypotheses of the study are:

Null Hypothesis (H_0)

Health indicators do not have a significant impact on Gross Domestic Product per capita.

$$H_0 : \beta_i = 0$$

Alternative Hypothesis (H_1)

Health indicators have a significant impact on Gross Domestic Product per capita.

$$H_1 : \beta_i \neq 0$$

where β_i represents the coefficient of each explanatory variable.

Expected relationships between variables and the dependent variable are as follows:

Variable	Expected Relationship
Life Expectancy	Positive
Current Health Expenditure	Positive
Domestic Private Health Expenditure	Positive
Government Expenditure	Positive
Under-5 Mortality Rate	Negative

The study expects that improvements in health conditions and health expenditure will positively influence economic growth, while higher child mortality will negatively affect economic performance.

Model Selection Hypotheses

To determine the appropriate estimation technique, pooled OLS and panel data models are compared.

Breusch–Pagan Lagrange Multiplier (LM) Test

H_0 : There are no panel effects and pooled OLS is appropriate.

H_1 : Panel effects exist and a panel data model (Fixed Effects / Random Effects) is preferred.

The LM test is first employed to determine whether panel estimation is required.

Hausman Specification Test

H₀: Random Effects model is appropriate and consistent.

H₁: Fixed Effects model is appropriate and random effect model is inconsistent

The Hausman test is conducted to select between the Fixed Effects and Random Effects estimators.

3. Data and Methodology

The study is based on a panel data set consisting of both cross-sectional (countries) and time-series (years) observations allowing the use of panel data techniques to capture variations across countries as well as over time. In this study a balanced panel of (13 × 21) 273 observations is used. The analysis includes selected South Asian and Southeast Asian economies over the period 2000–2020. The countries included in the study are India, Pakistan, Bangladesh, Nepal, Bhutan, Indonesia, Malaysia, Singapore, Thailand, Vietnam, the Philippines, Cambodia, and Brunei Darussalam. The selected period was chosen based on data availability and consistency across countries. The chosen time period covers the first two decades of the twenty-first century, a period characterized by rapid economic growth, increasing globalization, and substantial improvements in health outcomes across developing and emerging economies uplifting the underdeveloped regions of the world. The study restricts the sample to the pre-pandemic period in order to avoid the economic and social disruptions caused by the COVID-19 crisis, which could otherwise distort the empirical findings. These economies represent a diverse group of countries with varying levels of economic development, healthcare infrastructure, and public health performance. The selected countries have experienced significant structural transformation, rising income levels, and improvements in human development indicators during the sample period, making them suitable for examining the relationship between health and economic growth.

All data used in the study are secondary in nature and have been collected from the World Development Indicators (WDI) database published by the World Bank.

The variables used in the model are :-

- **Gross Domestic Product per capita (GDPPC)** is used as the dependent variable representing economic growth.
- **Life expectancy at birth (LEB)** and **under-5 mortality rate (U5MR)** are used as indicators of health outcomes.
- **Current health expenditure per capita (CHEPC)** and **domestic private health expenditure per capita (DPHE)** are included to represent healthcare spending.
- **General government expenditure (GGE)** as a percentage of GDP is used as a control variable.

These variables have been widely used in existing literature to analyze the relationship between health and economic growth. To reduce heteroscedasticity and avoid spurious regression

results, all variables are converted into natural logarithmic form. Logarithmic transformation also simplifies the interpretation of coefficients in percentage terms.

Indicator	Symbol	Units	Source
Gross Domestic Product per Capita	GDPPC	Constant 2015 US\$	World Bank
General Government Expenditure	GGE	% of GDP	World Bank
Domestic Private Health Expenditure per capita	DPHE	Current US\$	World Bank
Current Health Expenditure per Capita	CHEPC	Current US\$	World Bank
Life Expectancy at Birth	LEB	Years	World Bank
Under-5 Mortality Rate	U5MR	Per 1,000 live births	World Bank

To analyze the impact of health on economic growth empirically in South Asia and Southeast Asia from 2000 to 2020, a panel data model is used

$$Y_{it} = \beta_0 + \beta_j Z_{it} + U_{it}$$

i = 1 to 13 represents the countries' economies, t = 2000 to 2020 represents a time period of 21 years, U_{it} is the error term, β 's are the coefficients, Y is the dependent variable and Z is the vector of independent variables.

The model used in this study is given below-

$$LGDP_{PCit} = \beta_0 + \beta_1 LLEB_{it} + \beta_2 LU5MR_{it} + \beta_3 LCHEPC_{it} + \beta_4 LDPHE_{it} + \beta_5 LGGE_{it} + U_{it}$$

The dataset was first imported into Stata software and checked for consistency, missing values, and variable formats. Since the data consisted of multiple countries observed over several years, the country variable was encoded into a numerical identifier, and the panel structure was defined using country and year dimensions. All variables were then transformed into their natural logarithmic form to maintain consistency with the original study, reduce potential heteroscedasticity, and facilitate interpretation of the estimated coefficients.

The empirical analysis began with the estimation of a pooled Ordinary Least Squares (OLS) regression model to examine the relationship between health indicators and economic growth. Descriptive statistics were also generated to summarize the characteristics of the variables used in the study.

To determine whether panel estimation was more appropriate than pooled OLS, the Breusch–Pagan Lagrange Multiplier (LM) test was conducted. Based on the outcome of the LM test, panel data estimation techniques were subsequently employed. Both Fixed Effects (FE)

and Random Effects (RE) models were estimated to account for unobserved country-specific heterogeneity.

To select the most appropriate panel estimator, the Hausman specification test was performed. The test was used to determine whether the Fixed Effects model or the Random Effects model provided more consistent estimates. In addition, diagnostic tests were conducted to examine potential econometric issues such as multicollinearity and heteroskedasticity. The Variance Inflation Factor (VIF) was used to assess multicollinearity among explanatory variables and Breusch–Pagan test for heteroskedasticity.

4. Descriptive Analysis, Results and Interpretation

This section presents the empirical results and interpretation of the estimated models.

. sum

Variable	Obs	Mean	Std. Dev.	Min	Max
country	0				
year	273	2010	6.066422	2000	2020
gdppc	273	8516.455	14571.74	547.2211	61345.53
leb	273	70.51893	5.072598	59.491	83.59512
u5mr	273	35.82784	26.17813	2.4	107.9
chepc	273	256.1231	510.3825	8.100258	3541.365
dphe	273	114.1175	263.0672	3.287654	1419.46
gge	273	11.5357	5.33868	4.809339	29.57622
lngdp	273	8.037476	1.290351	6.304853	11.02428
lnleb	273	4.253337	.0712642	4.085825	4.425985
lnu5mr	273	3.229109	.9466174	.8754688	4.681205
lnchepc	273	4.471581	1.3622	2.091896	8.172268
lndphe	273	3.696684	1.224032	1.190174	7.258032
lngge	273	2.349471	.4337866	1.57056	3.386971

. vif

Variable	VIF	1/VIF
lnchepc	21.53	0.046454
lnu5mr	20.20	0.049517
lnleb	11.81	0.084694
lndphe	10.44	0.095815
lngge	3.11	0.321380
Mean VIF	13.42	

- A. Life expectancy (LEB) and Under-5 mortality (U5MR) are almost mirror images of health status. Current health expenditure (CHEPC) and Private health expenditure (DPHE) are closely related spending measures.

The **Variance Inflation Factor (VIF)** test indicated the presence of multicollinearity is economically sensible. This is expected because life expectancy and under-5 mortality both measure health outcomes, while current health expenditure and private health expenditure capture related aspects of healthcare spending. Since these variables are theoretically important and are consistent with the original study, they were retained in the analysis.

- B. We also check for heteroskedasticity in the model

Command - estat hettest.

Ho: Constant variance

chi2(1) = 1.66

Prob > chi2 = 0.1982

The Breusch–Pagan heteroskedasticity test yielded a p-value of 0.1982. Since the p-value exceeds 0.05, the null hypothesis of constant variance was not rejected, indicating the absence of heteroskedasticity in the model.

- C. The first step is to run Pooled OLS regression with my dependent variable and independent regressors.

Command - reg lngdp lnleb lnu5mr lnchepc lndphe lngge

lngdp	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lnleb	-5.544914	.9051404	-6.13	0.000	-7.327034	-3.762793
lnu5mr	-.8752173	.0891174	-9.82	0.000	-1.05068	-.6997551
lnchepc	.7338745	.063938	11.48	0.000	.6079877	.8597612
lndphe	-.2041015	.0495453	-4.12	0.000	-.3016507	-.1065523
lngge	.1722263	.0763356	2.26	0.025	.0219301	.3225225
_cons	31.51631	4.086845	7.71	0.000	23.46977	39.56286

However, Pooled OLS treats all observations as a single cross-sectional dataset and ignores the panel structure of the data. Different countries may possess unique characteristics. If such unobserved factors influence the dependent variable and are correlated with the explanatory variables, pooled OLS estimates may become biased and inconsistent due to omitted variable bias. Pooled OLS assumes that all countries are homogeneous and ignores country-specific characteristics and unobserved heterogeneity.

The pooled OLS results indicate that all explanatory variables are statistically significant at the 5% level, as their p-values are below 0.05. Although all variables are statistically significant, some coefficient signs are inconsistent with economic theory. Life expectancy and domestic private health expenditure are expected to positively influence economic growth, yet pooled OLS estimates negative relationships. This suggests that pooled OLS fails to adequately account for unobserved country-specific characteristics and may suffer from omitted variable bias. Therefore, despite the statistical significance of the coefficients, the pooled OLS model is not considered the most appropriate specification for the analysis. The problem with pooled OLS is not the significance but that some estimated relationships are economically implausible because country-specific heterogeneity is ignored.

So, after estimating the pooled regression, the next step is to determine whether panel-specific effects exist. To examine this, the Breusch–Pagan Lagrange Multiplier (LM) test is employed to determine whether panel data estimation was more appropriate than pooled OLS.

- D. We complete the panel data setup by mentioning the country as cross-sectional units and year as time series units resulting in a **strongly balanced panel** comprising observations for all countries from **2000 to 2020**, with no missing time periods.

Command - xtset countryid year

panel variable: countryid (strongly balanced)

time variable: year, 2000 to 2020

delta: 1 unit

- E. Breusch–Pagan Lagrange Multiplier (LM) test gives the result (done only after employing Random effect model that is shown later)

Command - xttest0

Test: Var(u) = 0

chibar2 (01) = 1375.55

Prob > chibar2 = 0.0000

The test produced a chibar^2 statistic of 1375.55 with a p-value of 0.0000. Since the p-value is less than 0.01, the null hypothesis of no panel effects is rejected. This indicates the presence of significant country-specific effects in the dataset and thus pooled regression was inappropriate.

- F. **Command - xtreg lngdp lnleb ln5mr lnchepc lndphe lngge, re**

lngdp	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
lnleb	.3063275	.5071153	0.60	0.546	-.6876003	1.300255
lnu5mr	-.519104	.0601008	-8.64	0.000	-.6368995	-.4013086
lnchepc	.0925398	.0310604	2.98	0.003	.0316626	.1534171
lndphe	.1107392	.0267692	4.14	0.000	.0582726	.1632059
lngge	.1099061	.0425776	2.58	0.010	.0264555	.1933567
_cons	7.329417	2.32468	3.15	0.002	2.773128	11.88571

The Random Effects estimation indicates that life expectancy has a positive coefficient ($\beta = 0.306$), suggesting that improvements in life expectancy contribute positively to economic growth. However, the relationship is statistically insignificant ($p = 0.546$), implying insufficient evidence to confirm its impact on GDP per capita. The under-five mortality rate exhibits a negative and highly significant effect on economic growth ($\beta = -0.519$, $p < 0.01$), indicating that higher child mortality adversely affects GDP per capita. Current health expenditure per capita shows a positive and statistically significant relationship with GDP per capita ($\beta = 0.093$, $p = 0.003$), suggesting that greater healthcare spending contributes to economic growth. Similarly, domestic private health expenditure per capita has a positive and highly significant effect ($\beta = 0.111$, $p < 0.01$), indicating that increased private investment in healthcare is associated with higher income levels. Government expenditure also exerts a positive and significant influence on GDP per capita ($\beta = 0.110$, $p = 0.010$).

Overall, the Random Effects results suggest that healthcare expenditure variables and government expenditure positively contribute to economic growth, while higher under-five mortality negatively affects economic performance. Life expectancy, although positive, does not exhibit a statistically significant impact in the model.

- G.** Now we do not know whether RE is the best model out there. So. to check whether we need a fixed effect model or not we need to employ the Hausman Test. But first we need to run the fixed model regression.

Command - xtreg lngdp lnleb lnu5mr lnchepc lndphe lngge, fe

lngdp	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lnleb	.9037524	.4791255	1.89	0.060	-.0397946	1.847299
lnu5mr	-.4331399	.0574233	-7.54	0.000	-.5462242	-.3200556
lnchepc	.0737529	.0288343	2.56	0.011	.0169692	.1305365
lndphe	.1335209	.0250013	5.34	0.000	.0842856	.1827561
lngge	.0880905	.0396899	2.22	0.027	.0099289	.1662522
_cons	4.561826	2.197508	2.08	0.039	.234251	8.889402

The Fixed Effects model indicates that life expectancy has a positive coefficient ($\beta = 0.904$), suggesting that improvements in life expectancy are associated with higher GDP per capita.

However, the relationship is only marginally significant at the 10 percent level ($p = 0.060$). The under-five mortality rate exhibits a negative and highly significant effect on economic growth ($\beta = -0.433$, $p < 0.01$), implying that higher child mortality adversely affects GDP per capita. Current health expenditure per capita has a positive and statistically significant impact on economic growth ($\beta = 0.074$, $p = 0.011$). Similarly, domestic private health expenditure per capita positively influences GDP per capita and is highly significant ($\beta = 0.134$, $p < 0.01$). Government expenditure also shows a positive and significant relationship with economic growth ($\beta = 0.088$, $p = 0.027$).

Overall, the Fixed Effects results suggest that healthcare expenditure and government expenditure contribute positively to economic growth, while under-five mortality has a detrimental effect on economic performance. Life expectancy displays a positive but only weakly significant relationship with GDP per capita.

- H. However we may notice that the FE model is very similar to the RE model and we cannot decide between the two just by looking at values. We might misassume that just because RE has a lower p-value, it's better. FE and RE answer different assumptions. We need to remember that random effects assumes $\text{Cov}(X_{it}, u_i) = 0$ i.e. country-specific effects are NOT correlated with explanatory variables. And Fixed Effects allows $\text{Cov}(X_{it}, u_i) \neq 0$ i.e. country-specific effects may be correlated with explanatory variables. Both models can give positive coefficient, significant p-value and a nice R^2 value. To decide more precisely we employ the Hausman test.

Command - hausman fe re

Test: Ho: difference in coefficients not systematic

$$\chi^2(5) = (b-B)'[(V_b - V_B)^{-1}](b-B)$$

= - 47.86 $\chi^2 < 0 \implies$ model fitted on these data fails to meet the asymptotic assumptions of the Hausman test.

The Hausman test result cannot be interpreted in the usual way because the test statistic is negative ($\chi^2 = -47.86$), which is theoretically impossible since a chi-square statistic must be ≥ 0 . This means the negative value indicates that the matrix $(b-B)'[(V_b - V_B)^{-1}](b-B)$ is not positive semidefinite, which violates a key assumption of the Hausman test. Because the test fails, you cannot conclude fixed vs random effects validity from this output.

- I. Now we use the same hausman command but with sigmamore. Sigmamore fixes the problem by making the matrix invertible and positive semidefinite (or closer to it). So instead of the unstable direct difference used in the standard Hausman test, it uses a more robust covariance structure, which prevents the negative chi-square problem.

Command - hausman fe re, sigmamore

Test: Ho: difference in coefficients not systematic

$$\begin{aligned}\chi^2(5) &= (b-B)'[(V_b-V_B)^{-1}](b-B) \\ &= 44.54\end{aligned}$$

Prob>chi2 = 0.0000

As $\chi^2(5) = 44.54$ i.e. positive it can be accepted and p-value = 0.0000 so we strongly reject the null hypothesis that the difference in coefficients is systematic.

This means that the Random Effects estimator is not consistent and the regressors are likely correlated with unobserved individual effects.

So, the Fixed Effects model is preferred.

5. Conclusion

The analysis conducted in this study using panel data techniques in order to account for country-specific heterogeneity that could not be captured through pooled regression.

The empirical analysis began with a pooled OLS model. Although the pooled regression indicated that all explanatory variables were statistically significant, some estimated relationships were inconsistent with economic theory. In particular, life expectancy and domestic private health expenditure exhibited negative coefficients, despite being expected to contribute positively to economic growth. This suggested the presence of unobserved country-specific effects that were not adequately captured by pooled OLS. The Breusch–Pagan Lagrange Multiplier test further confirmed the existence of significant panel effects, indicating that panel estimation techniques were more appropriate than pooled regression. Fixed Effects and Random Effects models were estimated and compared using the Hausman specification test. The Hausman test strongly rejected the Random Effects model, leading to the selection of the Fixed Effects model as the final specification.

The Fixed Effects results reveal that under-five mortality has a significant negative impact on GDP per capita, implying that poorer health outcomes among children adversely affect economic performance. Current health expenditure per capita, domestic private health expenditure per capita, and government expenditure all demonstrate positive and statistically significant effects on economic growth. These findings suggest that greater investment in healthcare and public expenditure contributes to improvements in economic performance by enhancing productivity, human capital formation, and overall population well-being. Life expectancy was found to have a positive coefficient, indicating that improvements in longevity are associated with higher income levels. However, the variable was only marginally significant in the final model.

Overall, the findings support the view that health is an important determinant of economic growth. Improvements in healthcare spending and reductions in child mortality contribute positively to economic development, highlighting the role of health investments in promoting

long-run growth. The results therefore emphasize the importance of strengthening healthcare systems, improving access to medical services, and increasing both public and private health expenditure in developing and emerging economies.

Comparison with the Original Study

The findings of this study are broadly consistent with those reported in the original paper by Raina and Dar (2025). Both studies conclude that health indicators play an important role in explaining economic growth and that improvements in health outcomes are associated with higher levels of GDP per capita. In particular, both studies find that healthcare expenditure contributes positively to economic performance, while higher under-five mortality negatively affects economic growth. Thus, the overall conclusion that better health conditions promote economic development remains unchanged across the two studies. Both studies arrived at a similar econometric conclusion. The panel nature of the data proved important in both cases, and panel estimation techniques were preferred over pooled regression. In the present study, the LM test rejected pooled OLS and the Hausman test supported the Fixed Effects model, indicating that unobserved country-specific characteristics significantly influence economic growth. This suggests that accounting for heterogeneity across countries is essential when examining the relationship between health and economic development.

In contrast, the original study did not report severe multicollinearity problems. One possible reason is the difference in sample composition. The original paper focused exclusively on South Asian countries, which are relatively similar in terms of economic development, healthcare systems, and demographic characteristics. The present study, however, combines countries from South Asia and Southeast Asia creating greater variation in income levels and health outcomes. For example, highly developed economies such as Singapore coexist with developing economies such as Nepal, Bangladesh, and Cambodia within the same sample. These substantial differences may strengthen the correlations among health-related variables and increase the likelihood of multicollinearity.

With respect to heteroskedasticity, the Breusch–Pagan test indicated no evidence of heteroskedasticity in the model, implying that the variance of the error term remains reasonably constant across observations. This supports the reliability of the estimated coefficients and statistical inferences. Similar to the original study, the diagnostic results suggest that the estimated panel model provides a suitable framework for examining the relationship between health indicators and economic growth.

In conclusion, although the sample composition differs from that of the original paper, the overall evidence remains consistent: investments in health and improvements in health outcomes contribute positively to economic growth, while poor health indicators such as child mortality hinder economic performance. The results reinforce the importance of health as a key component of sustainable economic development.